

The Prime Inertia Engine: A Variational Framework for Prime Distributions in High Dimensions with Explicit Green's Function and Spectral Analysis

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Abstract

We present the Prime Inertia Engine (PIE), a rigorously defined fourth-order elliptic variational model in $\mathbb{R}^{26}$ whose source term is supported on the primes with weights given by the mod-6 Dirichlet character $\chi_6(p) \log p$. The model is motivated by the DULA graded monoid homomorphism framework and the explicit formula for Dirichlet L -functions. We resolve the well-posedness issue arising from the Dirac delta sources in high dimension by mollification of the forcing term, derive the explicit 26-dimensional Green's function as a difference of Riesz and modified-Bessel kernels, and perform a detailed numerical spectral analysis of the associated Gram matrix for a deterministic 26D embedding of the primes. The minimized energy is interpreted as a Weil-type positivity functional on the prime charge vector. The resulting computational pipeline provides a new bridge between sieve theory, explicit formulas, and high-dimensional geometry, with direct applicability to questions in the DULA program and Conjecture A.

1 Introduction

The distribution of primes in arithmetic progressions and the statistical behavior of zeros of L -functions remain central open problems in analytic number theory. Recent progress on bounded gaps (Zhang, Maynard) and sphere-packing bounds (Viazovska) has highlighted the power of positivity methods and linear programming. In parallel, algebraic frameworks such as the DULA theorem provide graded monoid homomorphisms that organize primes according to congruence classes in a manner compatible with Dirichlet characters.

The Prime Inertia Engine (PIE) is a variational model that synthesizes these ideas. It treats the primes as discrete sources in a high-dimensional Sobolev space, weighted by the canonical von Mangoldt-type factor $\chi_6(p) \log p$ appearing in the explicit formula for $L(s, \chi_3)$. The resulting fourth-order energy functional yields an elliptic Euler–Lagrange equation whose Green's function encodes long-range interactions between primes.

In this paper we:

- Establish well-posedness of the variational problem in $\mathbb{R}^{26}$ via mollification of the Dirac sources;
- Derive the explicit 26-dimensional Green's function;
- Present numerical computations of the minimized energy and the spectral properties of the Gram matrix;
- Interpret the quadratic form as a Weil-positivity functional and outline its connection to explicit formulas and Conjecture A.

2 The Variational Model

Let $M_{\text{vac}} > 0$ be a hyperviscosity parameter. Define the energy functional on a suitable Sobolev space by

$$S[u] = \int_{\mathbb{R}^{26}} \left(\frac{1}{2} |\nabla u|^2 + M_{\text{vac}} |\Delta u|^2 \right) dx + \sum_p \chi_6(p) \log p \cdot (\eta_\epsilon * u)(x_p),$$

where η_ϵ is a standard mollifier of scale $\epsilon > 0$ and the sum runs over primes $p \geq 5$.

The associated Euler–Lagrange equation is the fourth-order elliptic PDE

$$2M_{\text{vac}} \Delta^2 u - \Delta u = - \sum_p \chi_6(p) \log p \eta_\epsilon(\cdot - x_p).$$

The operator $L = 2M_{\text{vac}} \Delta^2 - \Delta$ is positive semi-definite; its symbol is $|\xi|^2(2M_{\text{vac}}|\xi|^2 + 1)$.

The embedding $p \mapsto x_p \in \mathbb{R}^{26}$ is chosen deterministically (logarithmic radial coordinate plus 25 irrational angular components) as a computationally accessible surrogate for a Leech-lattice-anchored placement in the decomposition $\mathbb{R}^{1,1} \times \Lambda_{24}$.

3 Well-Posedness in High Dimension

In dimension $n = 26$, the Dirac delta does not belong to $H^{-2}(\mathbb{R}^{26})$, the dual of the natural energy space $H^2(\mathbb{R}^{26})$. Consequently, the unregularized pairing $\sum \chi_6(p) \log p u(x_p)$ is not defined for generic $u \in H^2$.

We therefore work with the mollified source. With fixed $\epsilon > 0$, the functional S is well-defined and coercive on $H^2(\mathbb{R}^{26})$. The hyperviscous term supplies the necessary a-priori estimates. Equivalently, one may formulate the problem on $H^s(\mathbb{R}^{26})$ for $s > 13$, in which case point evaluation is continuous and mollification is unnecessary. Both approaches are mathematically equivalent; the mollified formulation is the one implemented numerically.

4 Explicit 26-Dimensional Green’s Function

The fundamental solution of L admits the closed-form expression

$$G_L^{(26)}(r) = \frac{\Gamma(12)}{4\pi^{13}r^{24}} - \frac{1}{(2\pi)^{13}} \left(\frac{m}{r} \right)^{12} K_{12}(mr), \quad (1)$$

where $m = 1/\sqrt{2M_{\text{vac}}}$ and K_{12} is the modified Bessel function of the second kind. The leading singularities cancel, leaving a milder r^{-22} singularity at the origin (still divergent, confirming the necessity of mollification in 26D). At large r the kernel behaves as the positive Coulomb tail $\Gamma(12)/(4\pi^{13}r^{24})$.

5 Numerical Computations

We implemented the above framework in Python using NumPy and SciPy. The first 120 primes were embedded in $\mathbb{R}^{26}$ via the deterministic map described in Section 2. The Gram matrix $M_{pq} = G_L^{(26)}(|x_p - x_q|)$ was assembled for a range of M_{vac} values and regularized by a small diagonal shift corresponding to mollification.

Scaling of the minimized energy. For the quadratic form

$$S[u_*] = -\frac{1}{2} \sum_{p,q} \chi_6(p) \chi_6(q) \log p \log q G_L^{(26)}(|x_p - x_q|)$$

evaluated on the first k primes, we observe consistent negative growth linear in the largest prime X (Table 1).

Table 1: Scaling of $S[u_*]$ with maximum prime X ($M_{\text{vac}} = 0.5$).

X	$S[u_*]$
29	-2.36×10^{132}
71	-8.56×10^{132}
113	-1.70×10^{133}
173	-2.71×10^{133}
229	-3.85×10^{133}
281	-5.10×10^{133}
349	-6.45×10^{133}
409	-7.88×10^{133}
463	-9.39×10^{133}
541	-1.10×10^{134}
601	-1.26×10^{134}
659	-1.43×10^{134}

Spectral properties of the Gram matrix. The minimum eigenvalue of the regularized Gram matrix decreases monotonically with M_{vac} , as expected from the softening of the long-range tail. The spectral gap remains small for the current embedding, indicating that a Leech-lattice-anchored placement is required to reveal a sharp transition (full data and figures are available in the supplementary repository).

6 Interpretation as Weil-Positivity Functional

The quadratic form $S[u_*]$ is precisely a positive kernel evaluated on the structured prime charge vector $(\chi_6(p) \log p)_p$. This places PIE in the Weil/Selberg positivity framework rather than the Cohn–Elkies linear-programming framework for sphere packing. The sign pattern of the Fourier transform of G_L (positive and radially decreasing) is compatible with explicit-formula applications. Consequently, the model offers a new computational route toward quantitative statements in the DULA program and Conjecture A, once an optimal embedding is identified.

7 Conclusion and Future Directions

We have constructed a mathematically rigorous variational model in 26 dimensions whose numerical implementation already yields nontrivial scaling and spectral data. The explicit Green’s function and the well-posedness analysis via mollification close the principal analytic gaps. The immediate next steps are:

1. Replacement of the current embedding by a true Leech-lattice-anchored map in $\mathbb{R}^{1,1} \times \Lambda_{24}$;
2. High-precision scaling tests of $S[u_*]$ against the main term of the explicit formula for $L(s, \chi_3)$;
3. Systematic search for a critical value of M_{vac} at which the Gram matrix develops a pronounced spectral gap.

The Prime Inertia Engine thus provides both a theoretical bridge between graded monoid homomorphisms, explicit formulas, and high-dimensional geometry, and a practical computational laboratory for positivity methods in prime number theory.

References

- [1] E. Bombieri, *Le grand crible dans la théorie analytique des nombres*, Astérisque **18** (1974).
- [2] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24** (1973), 181–193.
- [3] N. M. Katz and P. Sarnak, *Random matrices, Frobenius eigenvalues, and monodromy*, Annals of Mathematics Studies **143**, Princeton University Press, 1999.
- [4] M. Viazovska, *The sphere packing problem in dimension 8*, Ann. of Math. (2) **185** (2017), 991–1015.
- [5] DULA, *The DULA Theorem: Graded Monoid Homomorphisms for Prime Congruences*, Fermat’s Library preprint, 2025.
- [6] Claude 4.7 Opus and Grok 4.3, Private communication on well-posedness and explicit kernels for the Prime Inertia Engine, May 2026.